

Kindle Agent - Product Requirements Document (PRD)

1. Product Overview

Kindle Agent is an AI-powered document conversion system that converts uploaded PDF documents into Kindle-compatible ebook formats through an agentic workflow.

The system automatically analyzes the uploaded PDF, determines the required processing steps, invokes appropriate tools, and returns a Kindle-ready file.

2. Problem Statement

Converting PDFs to Kindle-compatible formats often requires multiple manual steps:

- Determining whether OCR is required
- Extracting document metadata
- Cleaning formatting issues
- Converting to EPUB/AZW3
- Transferring files to Kindle

Users must currently operate multiple tools manually.

3. Goals

Primary Goals

- Upload a PDF
- Automatically determine processing requirements
- Produce a Kindle-compatible output
- Minimize user intervention

Secondary Goals

- Demonstrate agentic workflows
 - Learn MCP architecture
 - Build reusable document-processing tools
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4. User Flow

1. User uploads PDF
 2. Agent analyzes document
 3. Agent determines processing path
 4. Agent invokes required tools
 5. Agent generates Kindle-compatible file
 6. User downloads output
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5. Functional Requirements

PDF Analysis

The system shall:

- Detect whether a PDF is scanned or text-based
- Extract document metadata
- Determine document structure

OCR Processing

The system shall:

- Run OCR when required
- Skip OCR for text-based PDFs

Format Conversion

The system shall:

- Convert processed documents to EPUB
- Convert EPUB to AZW3
- Validate generated files

Metadata Management

The system shall:

- Extract title
- Extract author
- Allow metadata correction

Output Delivery

The system shall:

- Provide downloadable AZW3 file
 - Store generated outputs temporarily
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6. Non-Functional Requirements

Performance

- Process standard books within 2 minutes

Reliability

- Handle invalid PDFs gracefully
- Recover from tool failures

Usability

- Single upload interface
 - No manual workflow configuration
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7. Future Features

- Send directly to Kindle
- Multi-format support
- Batch processing
- EPUB enhancement using LLMs
- Reading-order correction